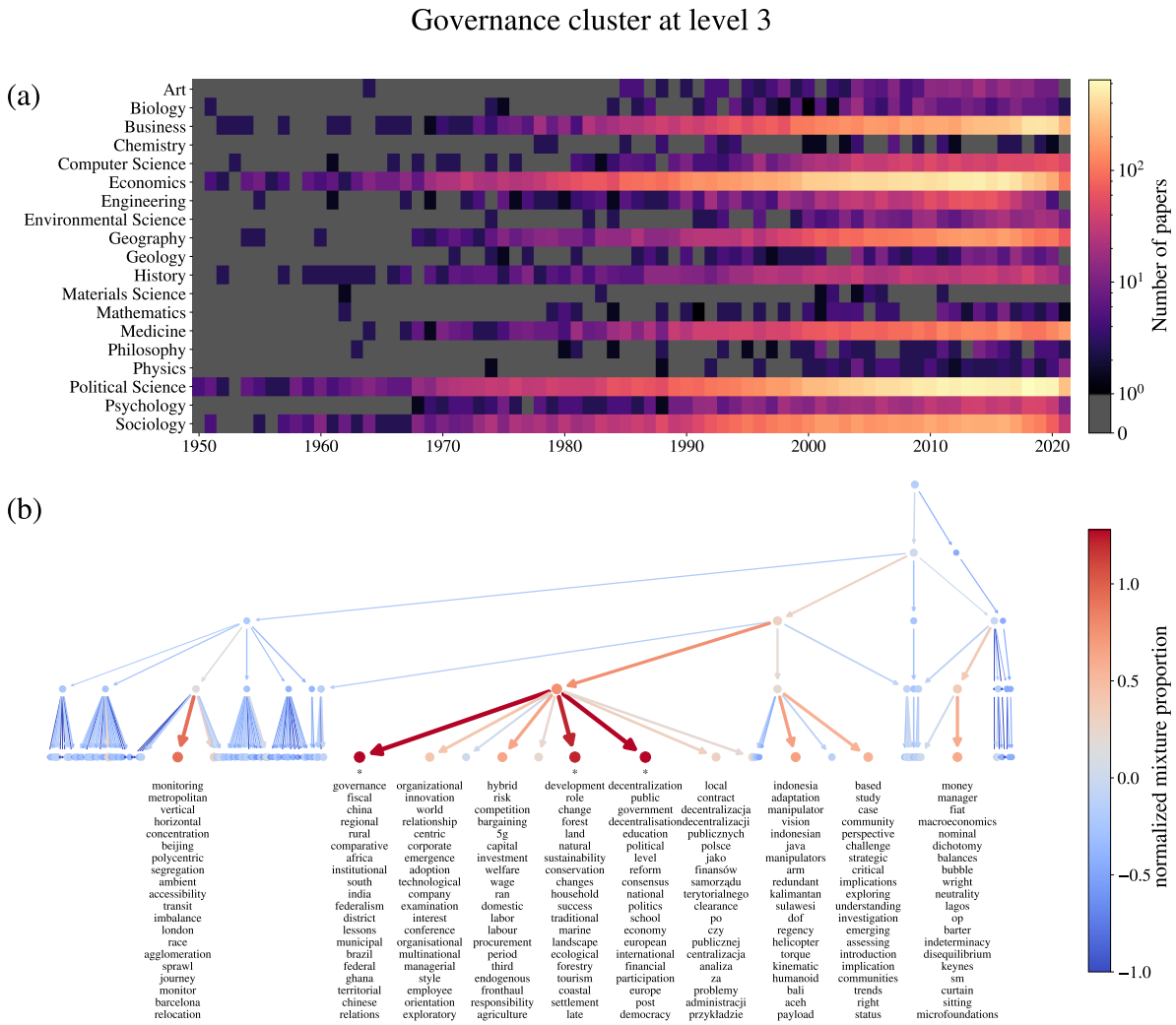


Figure S4: **Keywords annotation for Governance.** **(a)** Number of papers in time in each field of study for the Governance cluster. **(b)** Hierarchical topic tree, highlighting the top 20 words of the 10 most significant topics in the Governance cluster according to the normalized mixture proportion (in color). This information is used to aid the keywords annotation manual procedure.

| | keyword | count | sub-keywords | count |
|---|---|---|---|---|
| 0 | Robot swarms | 5 843 | routing, allocation, congestion | 909 |
| | | | robot swarms, supply chain | 4 120 |
| | | | planning, game theory | 693 |
| 1 | Cybersecurity | 18 437 | cybersecurity, peer2peer, fault tolearance, attacks | 15 189 |
| | | | cloud, security | 3 248 |
| 2 | Smart grids | 867 | smart grids, energy trading, blockchain | 867 |
| 3 | Blockchain | 9 637 | blockchain, cryptocurrency, ethereum, bitcoin | 9 634 |
| 4 | Federated learning | 2 043 | federated learning, deep learning, adversarial networks | 1 565 |
| 5 | Optimization | 5 878 | control theory, equilibrium | 926 |
| | | | statistical learning, optimization, detection | 3 065 |
| | | | heterogeneity, links, web, health | 1 887 |
| 6 | Control theory | 19 758 | control theory, nonlinear dynamics | 8 634 |
| | | | navigation, flocking, formation | 9 661 |
| | | | discrete-event systems, decision making | 1 463 |
| 7 | Electricity | 8 090 | electricity, grids | 8 090 |
| 8 | Investments | 8 394 | networks, connectivity, synchronization, routing, topology | 3 058 |
| | | | investments, money, risk, algorithm | 5 336 |
| 9 | Edge-computing | 2 970 | edge-computing, cloud-computing | 2 970 |
| 10 | Telecommunication | 18 804 | cellular networks, communication, radio | 12 446 |
| | | | wireless, localization | 1 262 |
| | | | routing, protocols, security | 5 096 |
| 11 | Wireless technologies | 2 166 | spanish | 1 251 |
| | | | wireless, cybersecurity, allocation, decision making | 915 |
| 12 | Governance | 40 045 | governance, fiscal federalism, government, development | 19 371 |
| | | | natural resources, education, governance | 20 674 |
| 13 | Environment | 15 386 | routing | 1 278 |
| | | | renewable energy, wastewater, environment | 8 348 |
| | | | portoguese, latin america | 5 760 |
| 14 | Social network analysis | 10 375 | supply chain, manufacturing, pricing | 3 938 |
| | | | social network analysis, organizations, firms | 6 437 |
| 15 | Health | 12 343 | algebra, healthcare, symptoms | 6 291 |
| | | | hospitals, HIV, cancer, surgery | 6 052 |

Table S4: **Results of the keywords annotation procedure.** Clusters at 3$^{rd}$ level, with associated manually annotated keyword and number of publications, together with the same information for each of the clusters at hierarchical level 4 in its branch of the hierarchy. Single horizontal lines denote clusters at the 2$^{nd}$ hiearchical level, while the double horizontal lines denote the three branches at the 1$^{st}$ level. Clusters with less than 500 documents have been disregarded.

## S2.3 Results of the hSBM

In this section we report some additional visualizations of the results of the hSBM algorithm on the *(de)centralization* dataset. In Fig. S5 we show the number of papers in each cluster at hierarchical level 4, together with the hierarchical tree and the manually annotated keywords. In Fig. S6 we show the bipartite network of documents and words, together with the results of the hSBM algorithm and the manual annotation procedure. This figure shows how the various clusters are represented in terms of topic frequency. Nodes on the left, indeed, represent documents, while those on the right title words. Notice that, to proper represent topic frequencies, multiple instances of the same word are considered, one for each document the word is in. Links are colored based on the doc cluster they start from. On top of the bipartite network between docs and words, we show a tree where squares represent clusters/topics at various hierarchical levels as a result of the hSBM, starting from the common root to level 3.

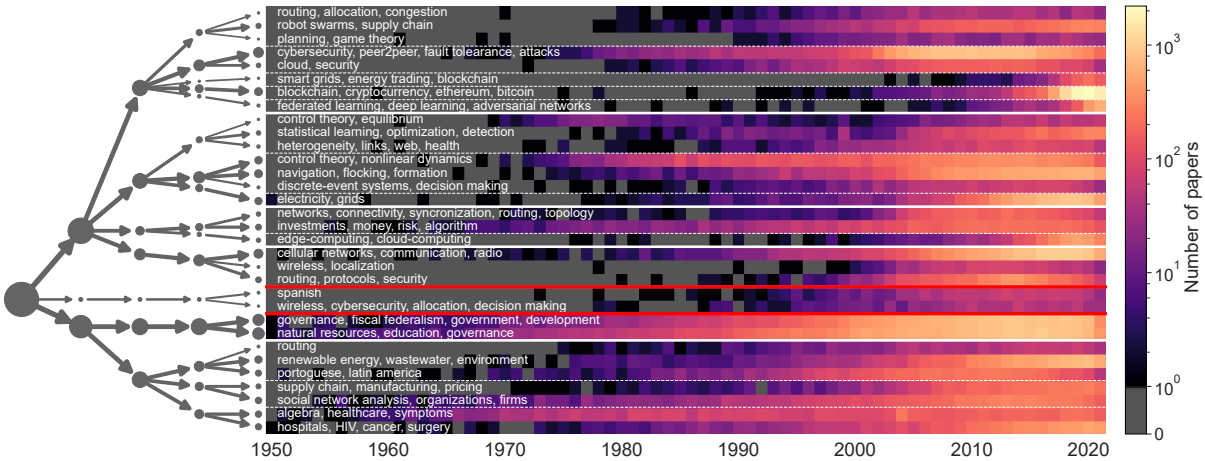


Figure S5: **Results of hSBM at the 4$^{\text{th}}$ hierarchical level.** Heatmap of the number of papers in each cluster at level 4, together with the manually annotated keywords and the hierarchical tree resulting from the clustering algorithm.